Reflection Paper: Car Price Prediction Using Regression Models

1. What Did I Implement?

In this assignment, I implemented a machine learning project to predict car prices using the cleaned dataset produced in Assignment Three (clean_car_dataset.csv). The dataset contained several features related to cars, such as odometer reading, number of doors, accident history, manufacturing year, location information, car age, and kilometers driven per year.

First, I loaded the cleaned dataset into a Jupyter Notebook using the Pandas library. The target variable (y) was the **Price** column, while the feature set (X) included all remaining columns except **Price** and **LogPrice**. The dataset was then divided into training and testing sets using an 80:20 ratio with random_state=42 to ensure reproducibility.

Two machine learning models were trained: **Linear Regression** and **Random Forest Regressor**. Linear Regression was used as a baseline model because it is simple and assumes a linear relationship between the input features and car prices. Random Forest Regressor was trained with 100 decision trees (n_estimators=100) to capture more complex relationships within the data. After training, both models were evaluated using performance metrics including R^2, MAE, MSE, and RMSE.

2. Comparison of Models

During the sanity check, predictions from the two models showed some differences. The Linear Regression model generally produced predictions based on a linear relationship between the features and the target variable. In some cases, its predictions were either slightly higher or lower than the actual car prices because it could not fully capture complex patterns in the dataset.

The Random Forest model usually produced predictions that were closer to the actual prices. Since Random Forest can learn non-linear relationships and interactions among variables, its estimates appeared more realistic. In my implementation, the Random Forest predictions better reflected the true market value of the cars, especially when several factors jointly influenced the price.

3. Understanding Random Forest

Random Forest is an ensemble machine learning algorithm that combines many decision trees to make predictions. Instead of relying on a single decision tree, the algorithm creates multiple trees using random subsets of the training data and random subsets of features.

Each decision tree independently predicts a car price. The final prediction of the Random Forest model is obtained by averaging the predictions from all trees. This ensemble approach reduces the risk of overfitting and generally improves prediction accuracy compared to using a single decision tree.

Because many trees contribute to the final prediction, Random Forest is usually more robust, stable, and accurate when dealing with complex datasets.

4. Metrics Discussion

Based on the evaluation results, the Random Forest model achieved a higher **R^2 score** and lower error metrics (**MAE** and **RMSE**) compared to Linear Regression. A higher R^2 value indicates that the model explains a larger proportion of the variation in car prices. Lower MAE and RMSE values indicate that the model's predictions are closer to the actual prices.

These results suggest that Random Forest is better at capturing complex relationships among features. On the other hand, Linear Regression is easier to interpret and computationally efficient, but its performance decreases when the relationship between variables is not purely linear.

Therefore, Linear Regression serves as a useful baseline model, while Random Forest provides improved predictive performance for this dataset.

5. My Findings

Overall, I prefer the Random Forest model for car price prediction. The evaluation metrics demonstrated that it produced more accurate predictions and lower prediction errors than Linear Regression. In addition, the sanity check showed that Random Forest generated predictions that were closer to the actual car prices.

The main reason for preferring Random Forest is its ability to model non-linear relationships and interactions among multiple variables. Car prices are influenced by many factors simultaneously, such as age, mileage, accident history, and location. Random Forest can effectively capture these complex relationships, making it a more suitable model for real-world car price prediction tasks. However, Linear Regression remains valuable because it is simple, interpretable, and computationally efficient.